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Determination of NIR informative wavebands for transmission non-invasive blood glucose measurement using a Fourier transform spectrometer

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Non-invasive blood glucose measurement using near infrared (NIR) spectroscopy relies on wavebands that provide reliable information about spectral absorption. In this study, we investigated wavebands which are informative for blood glucose in the NIR shortwave band (900~1450 nm) and the first overtone band (1450~1700 nm) through a specially designed NIR Fourier transform spectrometer (FTS), which featured a test fixture (where a sample or subject's finger could be placed) and all-reflective optics, except for a Michelson structure. Different concentrations of glucose solution and seven volunteers who had undergone oral glucose tolerance tests (OGTT) were studied to acquire transmission spectra in the shortwave band and the first overtone band. Characteristic peaks of glucose absorption were identified from the spectra of glucose aqueous solution by second-order derivative processing. The wavebands linked to blood glucose were successfully estimated through spectra of the middle fingertip of OGTT participants by a simple linear regression and correlation coefficient. The light intensity difference showed that glucose absorption in the first overtone band was much more prominent than it was in the shortwave band. *The results of the SLR model established from seven OGTTs in total on seven participants enabled a positive estimation of the glucose-linked wavelength*. It is suggested that wavebands with prominent characteristic peaks, a high correlation coefficient between blood glucose and light intensity difference and a relatively low standard deviation of predicted values will be the most informative wavebands for transmission non-invasive blood glucose measurement methods. This work provides a guidance for waveband selection for the development of non-invasive NIR blood glucose measurement. © *2018 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (http://creativecommons.org/licenses/by/4.0/).*
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I. INTRODUCTION

The level of blood glucose, an important indicator of bodily heath, is usually detected in invasive ways, such as using a fingerstick. A reliable and precise method for non-invasive blood glucose detection will allow people to be free from the pain and potential infection caused by the extraction of blood through a lancing device, especially diabetics who need long-term and frequent measurement.[1,2] Because glucose has complex overtones and combination absorption in the near infrared (NIR), NIR spectroscopy has been investigated for more than 20 years for non-invasive blood glucose measurement by a number of companies and research laboratories.[3,4] Both transmission and reflective approaches have been applied to acquire NIR spectra through the finger, oral mucosa, lip or other parts of the body to obtain blood glucose absorption information.[5] However, the spectra are usually

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poor and uninformative for providing blood glucose information, because they are easily obscured by the absorption of water and other bodily components, such as hemoglobin, fats, and protein.[6,7] Furthermore, changes in the external environment, such as physical temperature, time, or machine drift or angle changes in spectral collection could seriously affect the reliability of the measurement.[8,9] Many researchers have attempted to improve the signal-to-noise ratio (SNR) of NIR spectra, the experimental setup and multivariate calibration algorithms. But it has been suggested that to solve these problems and to achieve NIR non-invasive blood glucose monitoring, an essential precondition is the identification of informative wavebands for blood glucose.[10,11]

Multivariate statistical analysis methods, such as partial least squares regression (PLSR), multiple linear regression (MLR), or principal component regression (PCR) have been applied to the calibration of a regression model linking NIR spectra and glucose concentration. Several variables from the NIR spectra do not contain useful information about glucose absorption. Usually, these variables are removed to simplify the calculation and to improve the robustness of the calibration.[12] Only a few informative variables are essential for the modeling. To acquire information about blood glucose absorption, many researchers have investigated the correlation between glucose and different NIR wavebands through transmission and reflective approaches. Uwadaira *et al.*[10] performed 391 two-hour carbohydrate tolerance tests using 34 participants to acquire a large data set of reflection spectra through the palm of the right-hand. They proposed that combination vibrations of CH at 1018 nm, the second overtone of the NH vibration at 1030 nm and the combination vibrations of CH at 1042 nm should be considered to be characteristic for NIR non-invasive blood glucose measurement. However, it has proven difficult to establish a precise correlation between these wavebands and blood glucose. Another recent report from Goodarzi *et al.* presented a method for selecting the most informative wavebands in the shortwave band (900~1450 nm), the first overtone band (1450~1700 nm) and the combination band (2000~2300 nm). With three sets of data collected from glucose in aqueous solution and human serum samples, they found that wavebands from both the first overtone band between 1600 and 1700 nm and the combination band from 2100 to 2300 nm were the most informative. However, NIR radiation is significantly scattered by human body tissues in these wavebands.[11] Different conclusions have been reported earlier. Marbach *et al.*[3] proposed the glucose absorbance maximum at 1575 nm in aqueous solution (100 mg/dL). Arnold *et al.*[13,14] reported that the most informative waveband was from 2174 nm to 2380 nm and used it to improve the PLS calibration. Maruo *et al.*[15] found the combination band from 2035nm to 2324 nm was the best selection for glucose, while Heise *et al.*[16] concluded that the combination wave ranges 1473~1831 nm and 2111~2374 nm were the optimal for glucose measurement. The same wavebands have also been reported elsewhere in the literature.[17] In the NIR wave range, the shortwave band and first overtone band can penetrate physical tissues more easily than the combination band. In a comparison of transmission and reflective methods in glucose determination using NIR spectroscopy, the transmission method is preferable for glucose monitoring.[18] Most researchers agree with the use of combination band (from 2050 nm to 2300 nm), but disagree about whether the shortwave band or the first overtone band is preferable.

In this study, we aimed to identify the most informative NIR wavebands in the shortwave band and the first overtone band for transmission non-invasive blood glucose measurement. We designed a Fourier transform spectrometer (FTS) with all-reflective optics except for a Michelson structure and a test fixture (where a sample or a subject's finger can be placed). Two sets of experiments were conducted to acquire transmission spectra from samples of glucose solution and from the middle fingertip of participants who had undergone oral glucose tolerance tests (OGTT). Characteristic peaks of glucose absorption and the correlation between blood glucose concentration and light intensity difference were provided for the various spectra obtained, enabling the wavebands linked to blood glucose to be discriminated for a transmission measurement method.

II. MATERIALS AND METHODS

A. Design of measurement system

It is difficult to acquire transmission spectra with high SNR from the fingertip using a traditional NIR spectrometer, such as a fiber or grating based spectrometer, because of its low throughput of

light radiation. A spectrometer with excellent throughput efficiency and high spectral resolution is essential for our experiments. We designed a Fourier transform NIR spectrometer, which featured all-reflective optics except for a Michelson structure and a test fixture. Using all-reflective optics with an Offner relay lens, both axial chromatic and magnification chromatic aberrations could be eliminated. This also contributed to the elimination of wavenumber aliasing.[19] We designed the test fixture to ensure that the samples and fingertips can be optimally placed for measurement. Figure 1 shows a schematic diagram of the NIR spectrometer we designed. The apparatus mainly consists of a light source, a test fixture, an interferometer, an Offner relay lens, and a refrigerated InGaAs detector. The light source is a 10 W (maximum power) tungsten halogen lamp made by OSRAM and equipped with a circuit board. The pixel size of the InGaAs detector is 50 × 500 μm and the number of pixels is 256 × 1. The internal temperature of the detector is controlled by a programmable thermal electronic cooler and can be dropped to 20°C lower than the room temperature. The actual response of the detector is in the range of 900~1700nm. The working band of the spectrometer has a strong spectral performance over the shortwave band and the first overtone band.

The interferometer has a wedge-shaped Michelson structure, as shown in the insert in the bottom left of Fig. 1. By slightly tilting one of the mirrors (mirror M2 in Fig. 1), the image of M2 and the virtual image M1' (dotted line), which represents the image of mirror M1, can form a wedge with small angle α and they generate an equal-thickness interferogram. The interferogram is focused on the plane of the detector by the Offner relay lens. The chromatic aberration caused by the interferometer is estimated to be small, due to the quasi-all-reflective optics.

We designed the spectrometer to acquire spectral samples over the range 900~1700 nm. The wedge angle α was determined from the maximum optical path difference (OPD) $OPD_{\max}$, half of the interferogram dimension N, and the detector pixel spacing d_{px}. The maximum wedge angle α can be determined from the equation:

$$\alpha_{\max} = \tan^{-1}[OPD_{\max}/(2Nd_{px})], \tag{1}$$

Due to the symmetry of the interferogram, half of the interferogram dimension is half of N = 128 pixels. Based on the Nyquist Shannon sampling condition, 64 interference orders can be sampled. To obtain an interferogram generated by the shortest wavelength, which is 900 nm, the upper limit of the OPD is $OPD_{\max}$=900nm×64 = 57600nm. The pixel spacing d_{px} of the InGaAs detector we used was 50μm. Therefore, the maximum wedge angle was calculated to be $\alpha_{\max} = 0.26^{o}$. Because the shortest wavelength was 900 nm, the maximum wavenumber $v_{\max}$ is 11111.1 cm^{-1} and the wavenumber resolution was $\delta_v = v_{\max}/N$=86.8 cm^{-1}. In this case, the position of 1700 nm sample (wavenumber 5882.4 cm^{-1}) was near the 68th point in the spectrum dimension. We could easily calculate that the maximum number of samples over 900~1700 nm is 60 points (68th~128th point in the spectrum dimension). This is the case for maximum spectral resolution. However, when the spectral resolution is high (larger OPD), the SNR will be low.[20] In practice, we set the wedge angle at α = 0.13° to

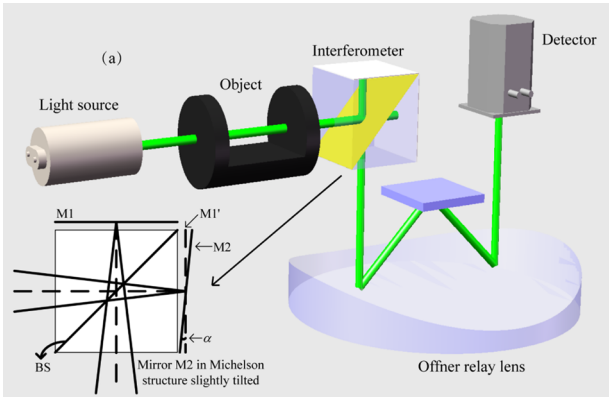


FIG. 1. Structural representation of the reflective interferential spectrometer. The insert on the bottom left shows the wedge-shaped Michelson interferometer.

achieve a better SNR and collected approximately 32 samples in the detector plane (39th~71th point in the spectrum dimension).

Our aim was to obtain NIR spectral curves with high SNR and high spectral resolution. Here, we present the spectral calibration used to determine the actual wavelength for each wavelength sample in the obtained spectral curve. The original interference data has one spatial dimension and contains a zero-frequency component and a cosine component. To eliminate the zero-frequency component, we used a self-adaptive differential filtering procedure, which was based on an iterative algorithm for dynamically adjusting its weighted mean filter window.[21] The correction formula with three-point mean value filtering is:

$$I_{cor}(x) = I(x) - r_n(x), \tag{2}$$

where $I(x)$ is the original interference curve, I_{cor} is the corrected curve and $r_n(x)$ is the residual of the zero-frequency component after n iterations. In our case, an optimal interference curve without zero-frequency component was achieved after seven iterations. Before the Fourier transformation, the interferogram was optimized by apodization.[22] Figure 2(a) shows the captured interferogram curve of the tungsten halogen lamp light source, while the zero-frequency elimination and apodization results are depicted in Fig. 2(b) and 2(c), respectively.

The spectrum was calculated by a spectral inversion algorithm based Fourier transform. The structure of the interferometer provides a linear distribution of the phase difference in resulting interferogram, and the wavenumbers of the spectral samples are also linearly distributed. To calibrate the wavelength position, we chose a 1625 nm NIR laser and a 1640 nm narrow band filter as the reference. However, the peak location of the reference spectrum was not always located in the center of a sampling pixel. Thus, we used Gaussian methods to subdivide the pixels. The vicinity of a peak in the reference spectrum can generally be approximated by a Gaussian distribution. The vertex of the fitted Gaussian distribution indicates the subpixel location of the spectral peak. The subpixel location of the monochromatic wavelength can be calculated:

$$n_{Gauss} = n - \frac{\ln S_{n+1} - \ln S_{n-1}}{2(\ln S_{n-1} - 2\ln S_n + \ln S_{n+1})}, \tag{3}$$

where S_{n+1}, S_n and S_{n-1} are the monochromatic amplitudes of the n+1th, nth, and n−1th pixels provided in the spectral curve. After matching the reference peak position and wavenumbers, the wavenumbers of the other sampling points were calculated through polynomial fitting. Figure 2(d) depicts the result of the calibration. Two prominent characteristic peaks were found at 1625.0 nm

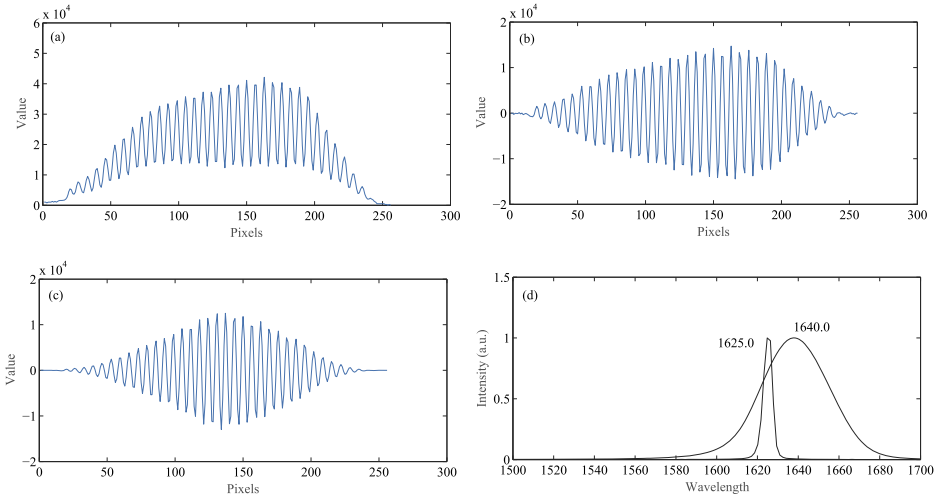


FIG. 2. Spectral calibration results. (a) Captured interferogram curve of the light source. (b) Zero-frequency elimination result. (c) Apodization result. (d) Spectra of 1625nm NIR laser and tungsten halogen lamp light source with a 1640nm narrow band filter.

and 1640.0 nm, corresponding to the spectrum references of the 1625 nm NIR laser light source and the 1640 nm narrow band filter, respectively. The calibration result then gives the correct spectra to the wavelength dimension of the spectrum data.

In summary, the spectrometer we have designed can provide precise wavelength samples for informative waveband determination of glucose over the range of the shortwave band and the first overtone band.

B. Experimental setup

Using the Fourier transform NIR spectrometer described above, two sets of experiments were designed for the investigation. The first set focused on glucose absorption in aqueous solution in the shortwave band and the first overtone band. We used a 1450nm edgepass filter to divide NIR light into two different ranges, the shortwave band (900~1450 nm) and the first overtone band (1450~1500 nm). According to Beer-Lambert law, in order to obtain more pronounced changes in light intensity, aqueous glucose solution with relatively higher concentration were chosen for experiment instead of concentration similar to actual blood glucose. One pure water sample and samples of different glucose aqueous solution (with concentrations ranging from 500 mg/dL to 3500 mg/dL, at intervals of 100 mg/dL) were prepared for the measurements. Samples were placed in a quartz cuvette (light path: 10 mm) one by one and were preheated to the same temperature of 37°C using a thermostatic water bath. The optical power of the light source for the solution samples was 4.7 W. The integration time of the detector was 80 ms in the shortwave band and 500 ms in the first overtone band. The spectra of each sample were recorded three times in the two different wavebands.

According to the glucose absorption waveband, the second experiment was performed to investigate the blood glucose linked wavebands in human subjects. We performed OGTT to obtain a reference blood glucose concentration. OGTT can provide a varying range of blood glucose concentrations over several hours. It has been used as a special experimental method for research into non-invasive blood glucose measurement.[10,23] Seven healthy male volunteers participated in this study. The ages of the volunteers were 23, 24, 24, 25, 25, 27, and 27 years and their body mass indexes were 23.28, 24.41, 21.38, 22.80, 24.98, 22.41, and 23.44 kg/m^2, respectively. Each of them drank 100 ml of water containing 75 g glucose in the morning after fasting for 8 hours overnight. NIR transmission spectra were collected from the middle fingertip in the shortwave band and the first overtone band. The optical power of the light source for the fingertips was 10 W. The integration time of the detector was 1500 ms in the shortwave band and was 4500 ms in the first overtone band. Conventional invasive blood glucose tests were conducted to obtain reference blood glucose concentrations by using a self-monitoring blood glucose meter (Acct-Chek® Active, Roche Diagnostics GMBH, Germany). Spectral acquisition and conventional blood glucose tests were performed simultaneously at 0 (fasting state), 30, 60, 90, 120,150, and 180 min after drinking the glucose water. The temperatures of each volunteer was measured to be 36.7±0.2°C, which would have little effect on the spectral accuracy of the blood glucose. In this experiment, the reference blood glucose had a mean concentration of 124.54 mg/dL (range: 75.6~178.2 mg/dL), which increased gradually to a peak value and then decreased. Each NIR transmission spectrum and reference blood glucose concentration was measured three times.

C. Determination of wavebands

The spectral value of each wavelength was preprocessed through a log transform:

$$D_\lambda = \log(R_\lambda), \tag{4}$$

in which R_λ is the value of the transmission spectra and λ is the wavelength. To eliminate spectral baseline shifts, a Savitzky–Golay filter with three-point smoothing[24] and second-order derivative processing were applied to discern the wavelength of the glucose absorption.[25] The derivative intensity was defined as follows:

$$I_{\lambda t} = \frac{d^2 D_\lambda}{d\lambda^2}, \tag{5}$$

where λ is the wavelength and t is the time after the volunteers had drunk. The value of $I_{\lambda t}$ represents the light intensity. It was used to calculate the light intensity difference $\Delta I_{\lambda t}$ between the drinking state and the fasting state as follows:

$$\Delta I_{\lambda t} = I_{\lambda t} - I_{\lambda 0}, \tag{6}$$

where $I_{\lambda 0}$ is the intensity at wavelength λ in the fasting state (t=0). To identify the wavebands linked to blood glucose, the correlation coefficient between the blood glucose concentration and the light intensity difference for each wavelength was calculated for every OGTT. Wavebands with the highest correlation coefficient provide a wavelength that is informative regarding blood glucose absorption. All data analyses were performed with MATLAB 2016a (MathWorks, Natick, MA, USA).

III. RESULTS AND DISCUSSION

A. Wavebands of glucose absorption in aqueous solution experiments

There were 105 sets of transmission spectra recorded in the shortwave band from glucose solutions, but only 35 sets of averaged spectral data are shown in Fig. 3(a), in which the curves were preprocessed using formula (4). The spectral curves arranged according to glucose concentrations represent the transmitted light intensity. The transmitted light intensity in the wave range 900~1450 nm is much higher than that in the wave range 1450~1700 nm. For the second derivative spectra, characteristic peaks in the shortwave band were observed at 938 nm, 1040 nm, 1135 nm, and 1295 nm, as depicted in Fig. 3(b). Because of the serious absorption of water, the change in light intensity around 1135 nm was small when the concentrations of the glucose solutions changed, but there was an obvious change in light intensity for the characteristic peaks of 938 nm, 1040 nm, and 1295 nm. This was caused by the different absorptions of varying concentrations of glucose solution. To emphasize the variation in the characteristic peaks of the spectra, five typical raw spectra collected from five different concentrations of glucose solution (500 mg/dL, 1000mg/dL, 1500 mg/dL, 2500 mg/dL, 3500 mg/dL) were selected for show in Fig. 3(c). With an increase in the concentration of the glucose solution, the transmitted light intensity decreased significantly in the broad wavebands 930~970 nm, 1040~1100 nm, and 1280~1300 nm. According to an earlier report, absorption caused by the CH_2 vibration occurs at approximately 930 nm, OH vibration was observed at approximately 970 nm and the combination vibrations of CH occur at approximately 1042 nm.[26] Characteristic peaks of glucose absorption in the shortwave band were confirmed from the spectra of the glucose solutions.

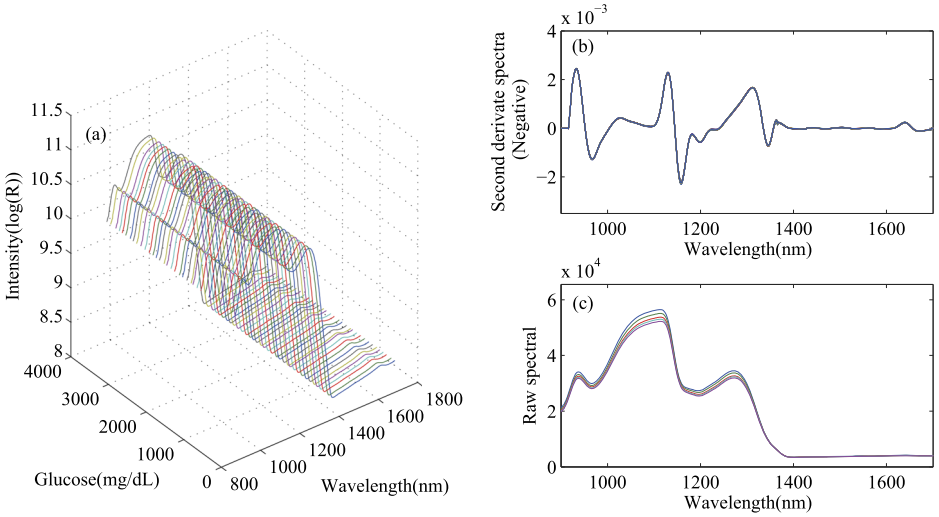


FIG. 3. Transmission spectra of pure water and glucose solution in the shortwave band. (a) Spectra arranged according to the glucose concentration, the curves were preprocessed using formula (4). (b) Second derivative spectra. (c) Five typical raw spectra with obvious absorption.

For the first overtone band, the waveband at 1616~1733 nm was reported to be the single most optimal informative waveband for glucose absorption.[27] And the first overtone band between 1600 nm and 1700 nm was concluded to have best predictive performance for continuous glucose monitoring in human serum.[11] The transmitted light intensity of the first overtone band was very low, which was attributed to a significant absorption by water. We applied a 1450 nm edgepass filter to suppress the shortwave band and to improve the spectral information of the first overtone band. There were 105 sets of transmission spectra recorded in the first overtone band, but only 35 sets of averaged spectral data are shown in Fig. 4(a), in which the curves were preprocessed using formula (4). The spectral curves arranged according to glucose concentrations represent the transmitted light intensity in the first overtone band. An obvious characteristic peak was observed around 1638nm from the raw spectra and the second derivative spectra are shown in Fig. 4(b). Unlike other wavebands that were seriously absorbed by water, a band with a width of nearly 40 nm (from 1610nm to1650nm) was only slightly absorbed by water. From the raw spectra in Fig. 4(a), it can be seen that the light intensity of this waveband clearly decreased when the concentration of the glucose solution increased. It is apparent in the five selected sets of spectra (obtained from concentrations of 500 mg/dL, 1000mg/dL, 1500 mg/dL, 2500 mg/dL, and 3500 mg/dL) depicted in Fig. 4(c), that there was a striking change in the peak values between the two adjacent spectra. Glucose absorption at around 1638 nm in the first overtone band was much more prominent than the absorption in the shortwave band. Experiment of aqueous glucose solutions with concentrations similar to actual blood glucose concentrations (80ml/dL, 100ml/dL, 150ml/dL, 200ml/dL) had also been performed in both shortwave band the first overtone band (data was not shown here). The level of light intensity changes was about one-fifth of the adopted concentration's (500-3500mg/dL). The change of light intensity was obvious, but slight. Wavebands of glucose absorption in both shortwave band and the first overtone band were visually distinguishable by experimental data acquired from the samples with the glucose concentrations (from 500 mg/dL to 3500 mg/dL, at intervals of 100 mg/dL).

B. Blood glucose linked wavebands for a transmission detection method

Figure 5(a) shows the transmitted light intensity of the left middle fingertip of each OGTT participant. The spectra curves were preprocessed using formula (4). Spectral curves with high intensity were recorded in the shortwave band and another set of spectral curves with low intensity were recorded in the first overtone band. There were 147 sets of spectra recorded in the shortwave band, with 49 sets of averaged spectral data are shown in Fig. 5(a). However only 21 sets of raw spectra were

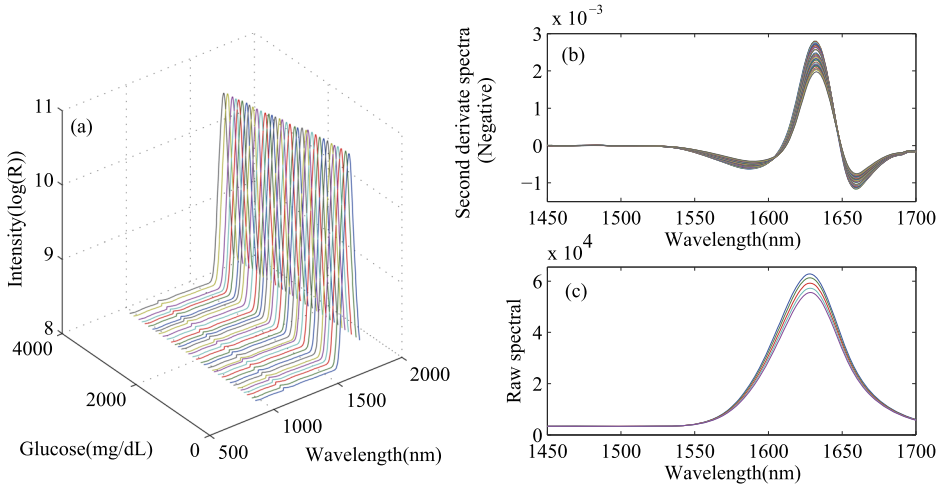


FIG. 4. Transmission spectra of pure water and glucose solution in the first overtone band. (a) Spectra arranged according to glucose concentration, the curves were preprocessed using formula (4). (b) Secondary derivative spectra. (c) Five typical raw spectra with an obvious absorption around 1638 nm.

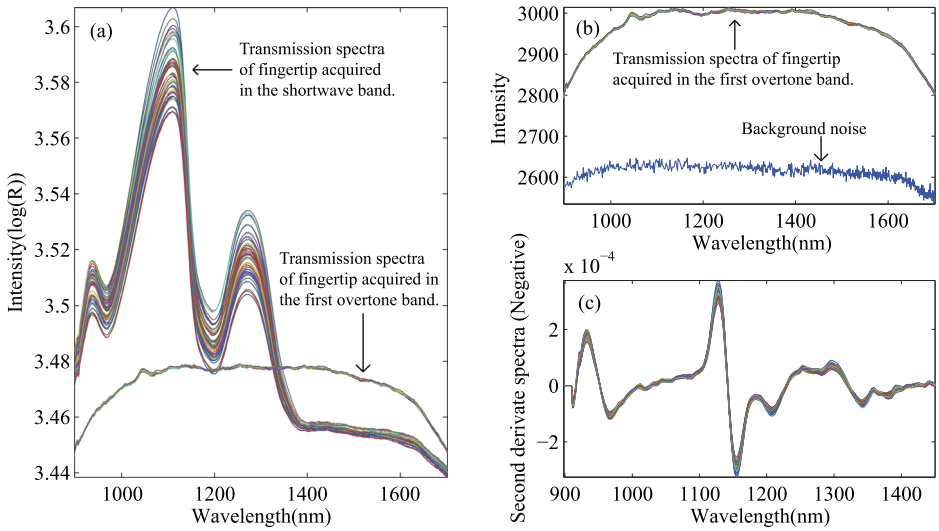


FIG. 5. (a) Transmission spectra of the left middle fingertip recorded in the wave ranges 900~1450 nm, and 1450 ~1700 nm, the curves were preprocessed using formula (4). (b) Raw spectra of left middle fingertip and background noise in the first overtone band. (c) Second derivative spectra of the left middle fingertip of oral glucose tolerance tests (OGTT) participants in the shortwave band.

recorded in the first overtone band due to its very low SNR. As shown in figure 5(b), when the optical power of the light source was 10 W (The maximum power is 10 W.) and the integration time of the detector was 4500 ms (the maximum integration time was limited to 5000 ms), the acquired values of the transmission spectra were slightly higher than the background noise. No useful information could be observed from the transmission spectra recorded in the first overtone band, whether the integration time was increased or the output power of the light source was improved during the experiment. The information from the spectra was almost obscured by noise. Under the experimental conditions, the prominent characteristic peaks of glucose absorption at 1600~1650 nm could not be observed in the spectra obtained from the middle fingertip through the transmission method. Consequently, the low transmittance of human tissue in the first overtone band resulted in it being unsuitable for transmission non-invasive blood glucose measurement.

In the shortwave band, the transmitted light energy was massively reduced because of reflection and scattering from body tissues. However, there were characteristic peaks that were consistent with the spectral characteristics of glucose concentration in the wavebands 930~970 nm, 1040~1100 nm, and 1280~1300 nm observed in the raw spectra in Fig. 5(a) and the second derivative spectra in Fig. 5(c). The light intensity of these characteristic peaks changed in the raw spectra of the middle fingertip, when the blood glucose concentration was varied in the OGTT. To obtain the informative wavelength for glucose absorption, the correlation coefficient (R_λ) for the relation between the reference blood glucose concentration (BG_t) and the light intensity difference ($\Delta I_{\lambda t} = I_{\lambda t} - I_{\lambda 0}$) was calculated for each OGTT. Figure 6 shows a histogram of the correlation coefficient R_λ, which had a range of 0.48~0.89, a mean value of 0.69 and a corresponding standard deviation of 0.13.

It is considered that a blood glucose linked wavelength is determined by a high correlation coefficient. In previous study,[10] when the correlation coefficient between the light intensity difference and the reference blood glucose concentration was greater than 0.7, the existence of a glucose-linked wavelength could be established. Here, we considered that when the correlation coefficient R_λ was greater than 0.72, the corresponding wavelength could provide information about the light intensity difference for the variation in spectral absorption. Table I summarizes four discerned broad wavebands with R_λ>0.72 and the corresponding peak wavelengths. These wavebands expressed as a percentage accounted for almost half of the shortwave band. High R_λ peaks were observed in the vicinity of 950 nm, 1150 nm, 1280 nm, and 1395nm (Table I). At these wavebands, prominent characteristic peaks were only observed in the wavebands 930~970 nm and 1280~1300 nm from the second-order spectra (Fig. 5(b)). The wavebands and the wavelengths of peak R_λ fluctuated in every participant.

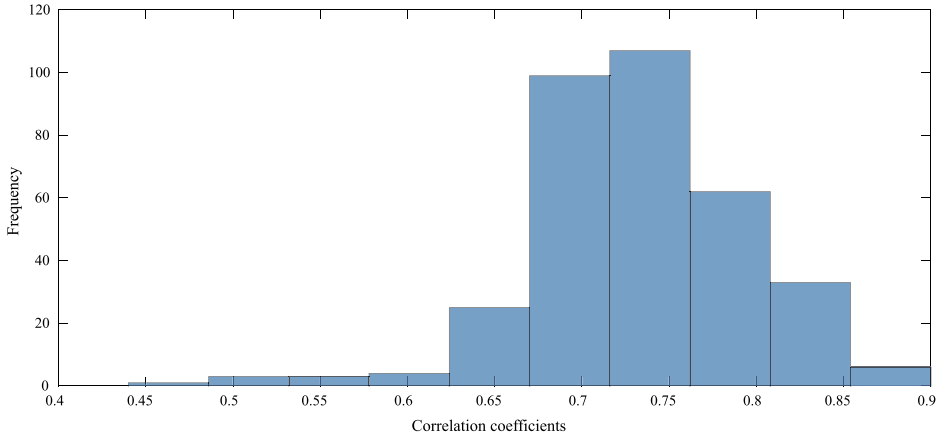


FIG. 6. Histogram of correlation coefficients between the reference blood glucose concentration and the light intensity difference for all OGTTs.

This could be due to the influence of external factors, such as biological tissue complexity, light source intensity drifts, and temperature variation.[28]

There were seven sets of transmission spectra and reference blood glucose concentrations for one OGTT. Each of the NIR transmission spectrum and the reference blood glucose concentration was recorded three times. Therefore, there were 21 data sets for each participant and total 147 data sets for all participant. The data sets were split into a calibration set of 98 samples (14 samples for each participant) and a test set of 49 samples (7 samples for each participant). Seven data sets for one OGTT were insufficient to establish a multivariate regression model for one participant, but they were sufficient to establish a simple line regression (SLR) model.[10] To identify the informative wavebands for blood glucose, we selected the two wavelengths 950 nm and 1280 nm to establish a model using SLR method between the reference blood glucose concentration and the light intensity difference for each participant. The correlation coefficient values of the selected wavelengths were 0.86 and 0.84, respectively, which were not the highest values among all the $R_\lambda > 0.72$ wavebands. The corresponding characteristic peaks were observed in the second derivative spectra. Figure 7 shows scatter plots for the reference blood glucose concentrations and the 49 predicted blood glucose concentrations at $\lambda = 950$ nm (filled circles) and 49 predicted blood glucose concentrations $\lambda = 1280$ nm (open circles). The mean standard deviation of predicted glucose concentration at wavelength 950 nm is 7.63 mg/dL and the mean standard deviation of predicted glucose concentration at wavelength 1280 nm is 13.25 mg/dL. A total of 91 predicted values (92.9%) were located in zone A which was an acceptable result, while the 7 predicted values (7.1%) in zone B had small errors and there were no values in zone C that would have significant errors.[29] Although the influenced factors would affect the fluctuation and the prediction accuracy, but they didn't affect an accepted prediction for the identification of informative waveband. The results indicated a positive estimation of the light intensity difference

TABLE I. Wavebands for which the correlation coefficients were greater than 0.72 for seven participants.

Participant	Par. 1	Par. 2	Par.3	Par.4	Par.5	Par.6	Par.7
Waveband R >0.72 (nm)	927~965 1135~1170 1267~1286 1379~1407	942~978 1151~1175 1259~1286 1387~1414	922~968 1141~1190 1268~1318 1388~1410	944~975 1057~1126 1257~1291 1388~1404	937~975 1049~1144 1256~1288 1385~1410	927~958 1079~1136 1257~1299 1387~1412	929~970 1070~1156 1243~1307 1380~1409
Wavelength of maximum R (nm)	950 1151 1273 1391	958 1167 1281 1401	957 1165 1280 1399	953 1113 1275 1395	957 1107 1286 1398	952 1122 1280 1399	955 1123 1281 1401

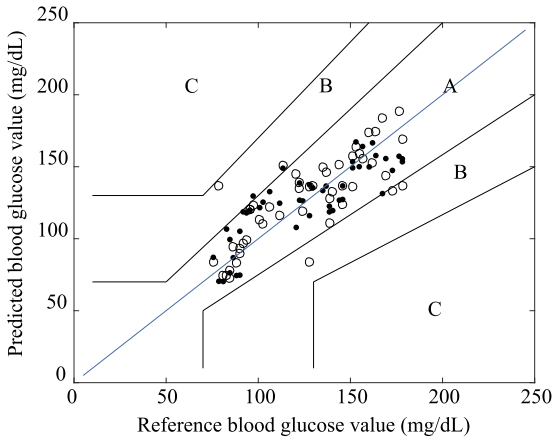


FIG. 7. Scatter plots of the reference blood concentrations and the predicted values calculated from a light intensity difference of λ = 950 nm (filled circles) and λ = 1280 nm (open circles). The average correlation coefficients for all participants were 0.82 and 0.79, respectively.

for the reference blood glucose concentration. The informative wavebands for blood glucose were identified through both the observed characteristic peaks from the spectra and their strong correlation. This study was an interesting basic investigation of non-invasive blood glucose measurement using NIR spectroscopy.

IV. CONCLUSION

In this study, we experimentally demonstrated an NIR FTS with a test fixture and all-reflective optics, except for a Michelson structure, to investigate the informative wavebands in the shortwave band and the first overtone band for transmission NIR blood glucose measurement. Transmission spectra were acquired through experiments with glucose solution and the middle fingertip of OGTT participants. Absorption by the glucose solution was observed in the shortwave band at 930~970 nm, 1040~1100 nm, 1280~1300 nm, and in the first overtone band at 1600~1650 nm. However, the characteristic peaks of the fingertip spectra were only observed in the shortwave band. This suggests that the NIR of the first overtone band is not capable of providing a transmission method of non-invasive glucose measurement. Furthermore, high correlation coefficients between the light intensity difference and the reference blood glucose concentration were obtained in several wavebands such as 950 nm and 1280 nm in the shortwave band. The results of the SLR model *established from seven OGTTs in total on seven participants also enabled* a positive estimation of the glucose-linked wavelength. This study identified an effective blood glucose linked wavelength to provide a reference for wavelength selection. Further work will concentrate on eliminating the effects of external factors and improving the accuracy of the light intensity difference for a single wavelength. More advanced detectors and filters will be introduced for the collection of light intensity.

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